

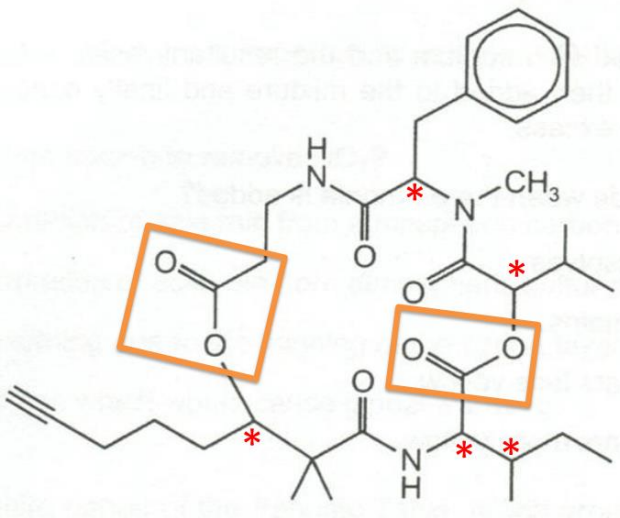
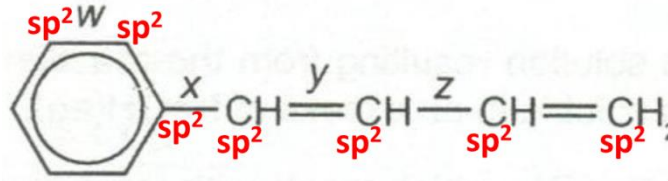
2013 9647 H2 Chemistry

Paper 1 Answer Key:

1	B	11	D	21	C	31	A
2	B	12	B	22	D	32	D
3	D	13	D	23	B	33	C
4	C	14	D	24	B	34	B
5	A	15	B	25	B	35	D
6	C	16	D	26	B	36	B
7	A	17	D	27	C	37	A
8	B	18	A	28	C	38	B
9	D	19	C	29	A	39	C
10	C	20	C	30	D	40	A

Paper 1 Worked Solutions

Q1 Answer: B $2 \text{H}_3\text{PO}_4 - \text{H}_2\text{O} = \text{H}_4\text{P}_2\text{O}_7$ $3 \text{H}_3\text{PO}_4 - 2\text{H}_2\text{O} = \text{H}_5\text{P}_3\text{O}_{10}$ Based on total number of H, P and O atoms of H_3PO_4 subtracting off the number of H and O atoms from water.	Q2 Answer: B <table><tr><th></th><th>Number of neutrons</th><th>Number of protons</th><th>Number of electrons</th></tr><tr><td>D_3O^+</td><td>11</td><td>11</td><td>10</td></tr><tr><td>H_3O^+</td><td>8</td><td>11</td><td>10</td></tr><tr><td>NH_2^-</td><td>8</td><td>9</td><td>10</td></tr><tr><td>OD^-</td><td>9</td><td>9</td><td>10</td></tr></table>		Number of neutrons	Number of protons	Number of electrons	D_3O^+	11	11	10	H_3O^+	8	11	10	NH_2^-	8	9	10	OD^-	9	9	10
	Number of neutrons	Number of protons	Number of electrons																		
D_3O^+	11	11	10																		
H_3O^+	8	11	10																		
NH_2^-	8	9	10																		
OD^-	9	9	10																		
Q3 Answer: D $\text{C}^+ : 1s^2 2s^2 2p^1 \rightarrow 1s^2 2s^2 2p^2$ $\text{N} : 1s^2 2s^2 2p^3 \rightarrow 1s^2 2s^2 2p^4$ $\text{Si}^- : 1s^2 2s^2 2p^6 3s^2 3p^3 \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^4$ $\text{P}^+ : 1s^2 2s^2 2p^6 3s^2 3p^2 \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^3$	Q4 Answer: C There is a large jump in the 4th IE to 5th IE, hence the 5th electron is removed from an inner shell and there are 4 valence electrons. Element M is from Group IV and hence it forms a chloride with a charge of +4.																				
Q5 Answer: A A compound with giant covalent structure will have high melting point and does not conduct electricity in any state.	Q6 Answer: C Gas behaves most ideally at <u>high temperature</u> and <u>low pressure</u>. - At <u>low</u> pressures, real gases behave <u>close to ideality</u> and $\frac{pV_m}{RT} \approx 1$. - This is because the gas particles are <u>widely spaced</u>, - Have <u>negligible volume compared to the volume of the container</u> and <u>negligible intermolecular forces of attraction</u> between the gas particles. - At <u>high temperatures</u> (keeping pressure constant), real gases behave <u>close to ideality</u> and deviation becomes less significant as $\frac{pV_m}{RT}$ approaches 1. This is because there are <u>negligible</u> intermolecular forces of attractions between the gas particles since they have <u>sufficient kinetic energy</u> to overcome the attractive forces.																				
Q7 Answer: A Using the formula: $\Delta H_{\text{rxn}}^\ominus = \Sigma \Delta H_f^\ominus (\text{products}) - \Sigma \Delta H_f^\ominus (\text{reactants})$ Hence, we need the $\Delta H_f(\text{CH}_3\text{CO}_2\text{Na}(\text{aq}))$, $\Delta H_f(\text{H}_2\text{O}(\text{l}))$ and $\Delta H_f(\text{CH}_3\text{CO}_2\text{Na} \cdot 3\text{H}_2\text{O}(\text{s}))$	Q8 Answer: B ΔG is negative because a vigorous reaction has taken place. ΔH is positive since the reaction is endothermic. $\Delta G = \Delta H - T\Delta S$ -ve = (+ve) - $T\Delta S$ ΔS must be negative.																				

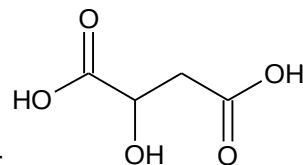
Q9 Answer: D Amount of Au = $\frac{6}{197}$ = 0.03045 mol $0.03045 = \frac{0.10 \times t}{3 \times 96500}$ t = 8.815×10^4 s	Q10 Answer: C IP = $0.10 \times 1 \times 10^{-9} \text{ } 1^2$ = $1 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$ Precipitation occurs when $IP > K_{sp}$. Only FeCO_3 and MnCO_3 are precipitated.										
Q11 Answer: D From table, it can be deduced that $\uparrow T$, $\uparrow K_w$, forward reaction favoured, forward reaction is endothermic. Therefore B and C are incorrect. At lower temperature, 0°C, backward reaction is favoured, equilibrium lies furthest to the left. A is incorrect as $[\text{H}^+]$ and $[\text{OH}^-]$ is equal for pure water at all temperatures. Since $\uparrow K_w$ with $\uparrow T$, $[\text{H}^+]$ increases and pH decreases. D is correct.	Q12 Answer: B NO_2 acts as a catalyst in the formation of acid rain from the oxidation of SO_2. The mechanism is as shown: $\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightarrow \text{NO}(\text{g}) + \text{SO}_3(\text{g})$ (slow) $\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{O}(\text{g})$ $\text{NO}(\text{g}) + \text{O}(\text{g}) \rightarrow \text{NO}_2(\text{g})$										
Q13 Answer: D The only property that decreases consistently (no anomalies) is radius of the atom.	Q14 Answer: D No of electrons contributed per atom: Mg (2) < Al(3) Electrical conductivity: Mg < Al Cu is a transition metal which has high electrical conductivity due to its available to contribute d electrons. Electrical conductivity: Al < Cu										
Q15 Answer: B Solution turns from pale green $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ to pale yellow $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ as MnO_4^- is added. Just beyond end-point, solution is pale pink due to small excess of MnO_4^-. Solution finally turns purple as more purple MnO_4^- in large excess.	Q16 Answer: D When $\text{CH}_2\text{Cl}/\text{CHICOOH}$ was heated in NaOH, Cl^- and I^- was released. When AgNO_3 was added, AgCl (white) and AgI (yellow) were formed. When conc ammonia was added, the white AgCl dissolved while yellow AgI the does not dissolved. The ppt appears more yellow.										
Q17 Answer: D Unpaired electron $\rightarrow$ Uneven number of electron <table border="1" data-bbox="97 1234 611 1395"> <tr> <th>Molecule</th><th></th></tr> <tr> <td>H_2S</td><td>18</td></tr> <tr> <td>Cl^+ and Cl^-</td><td>6 and 8</td></tr> <tr> <td>NH_4^+</td><td>8</td></tr> <tr> <td>Cu in CuO</td><td>27</td></tr> </table>	Molecule		H_2S	18	Cl^+ and Cl^-	6 and 8	NH_4^+	8	Cu in CuO	27	Q18 Answer: A A: X is Al. B: C and D: X is P.
Molecule											
H_2S	18										
Cl^+ and Cl^-	6 and 8										
NH_4^+	8										
Cu in CuO	27										
Q19 Answer: C  yanucamide B Ester linkage marked in box (2) Chiral centre marked in * (5)	Q20 Answer: C  All (w,x,y,and z) are made up of $sp^2 - sp^2$ overlap.										

Q21 Answer: C

Using the markovnikov's rule, iodine will be added to carbon 1 while chlorine will be added to carbon 2.

Q22 Answer: D

Step 1: Fumaric acid undergoes electrophilic addition to form the intermediate T, which contains a secondary alcohol.



Intermediate T:

Reagent required: steam and H_2SO_4

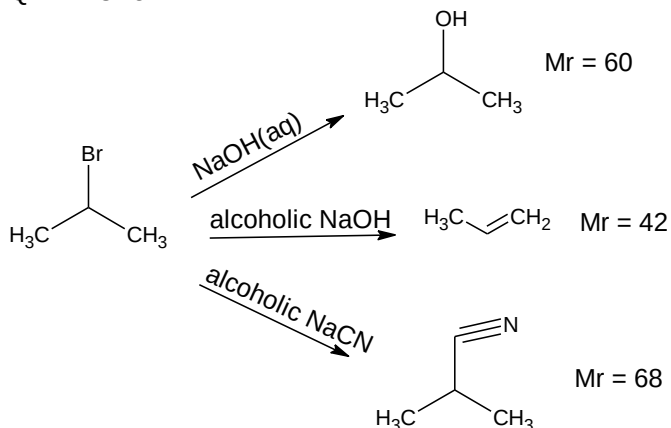
Step 2: The secondary alcohol functional group in intermediate T undergoes oxidation to form a ketone.

Reagent required: hot acidified KMnO_4

Q23 Answer: B

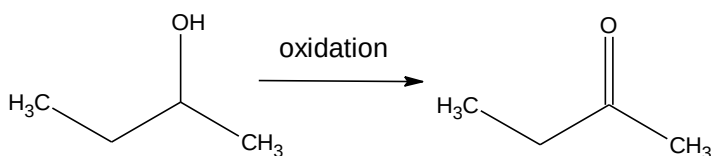
Option B contains two halogenoalkane functional groups. Hence it undergoes nucleophilic substitution with heated NaOH(aq) to form 2 primary alcohol functional groups. The Cl^- present in solution then reacts with acidified AgNO_3 to form AgCl , a white precipitate.

Q24 Answer: B



Q25 Answer: B

Butan-2-ol undergoes oxidation, forming a terminal methyl ketone which reacts positively with the iodoform test.



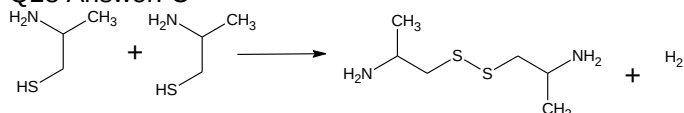
Q26 Answer: B

The 2 phenol functional groups in terbutaline reacts with NaOH to form a salt.

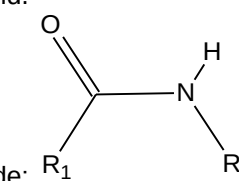
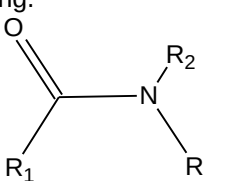
Q27 Answer: C

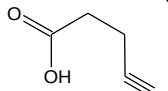
There are 3 moles of carboxylic acid and 1 alcohol functional group in citric acid. It reacts with 4 moles of PCl_5 , 4 moles of Na , and 3 moles of NaOH . It does not react with HCl .

Q28 Answer: C



The oxidation state of sulfur changes from -2 to -1. Therefore the formation of disulphide bridges is an oxidation reaction.

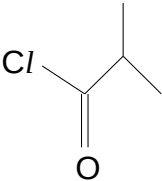
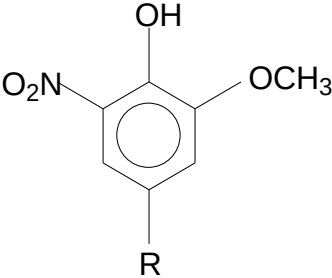
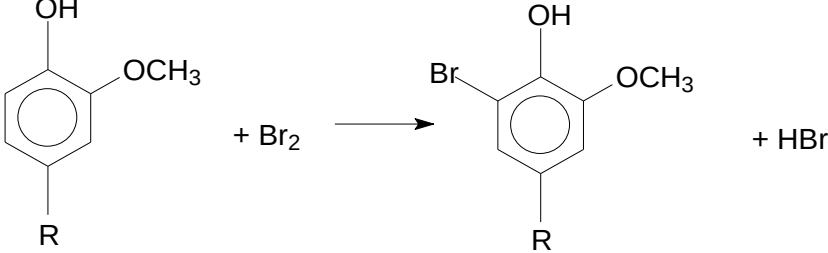
Q29 Answer: A Nucleophilic substitution occurs with the reactants in option A to form X.	Q30 Answer: D 'a bend in the regular shape' → hydrogen bonding to form the alpha helix structure. From option A to C, the a secondary amide is formed. This allows the <u>remaining hydrogen atom</u> bonded to N to <u>form hydrogen bonding</u> with the carbonyl carbon in the other amide bond.  Secondary amide: On the other hand, in option D, a tertiary amide is formed. There is <u>no hydrogen atom bonded to N</u> to form hydrogen bonding.  Tertiary amide:
Q31 Answer: A Upon complete combustion, 1 mole of diallyl sulfide produces 6 moles of CO₂. Amount of diallyl sulfide = 0.1/114 = 8.77 × 10⁻⁴ Amount of CO₂ = 21/24000 = 5.227 × 10⁻³ Ratio of diallyl sulfide to CO₂ = 1:6 Therefore, option 1 is correct. Upon complete combustion, 1 mole of diallyl sulfide produces 1 mole of SO₂. Volume of SO₂ = 8.77 × 10⁻⁴ × 24000 = 21 cm³ Therefore, option 2 is correct. When excess bromine is added, electrophilic addition occurs at the alkene functional group. As there are two alkene functional groups, there will be a total of 4 bromine atoms added to diallyl sulfide. Molecular mass of product = 114 + 4(79.9) = 433.6 Mass of product = 8.77 × 10⁻⁴ × 433.6 = 0.380 g Therefore, option 3 is correct.	Q32 Answer: D The bromine radical is the catalyst. Bromine and methane is the reactants.
Q33 Answer: C Option 1 is incorrect because there is <u>no need</u> to dry the gas. The hydrogen gas will be passed through the aqueous medium of H⁺. Option 2 and 3 are correct because the electrolyte must be 1 mol dm⁻³ and an inert electrode must be used for the standard hydrogen half cell.	Q34 Answer: B Option 1 is correct. M in MO₂ is oxidised to form MO. O in MO₂ is reduced to O₂. Option 2 is correct. The anion is O₂²⁻. Option 3 is incorrect. O²⁻ in MO has a smaller anionic radius than O₂²⁻. Therefore the lattice energy of MO should be greater than the lattice energy of MO₂. $ \Delta H_{\text{latt}} \propto \left \frac{q_+ q_-}{r_+ + r_-} \right $
Q35 Answer: D Option 1 is correct. Down the group, the ionic radius increases as the number of quantum shells increases. Option 2 is incorrect because $\Delta H_{\text{hyd}}(\text{cation}) \propto \left \frac{q_+}{r_+} \right$. Therefore down the group, the magnitude of hydration energy should decrease down the group. Option 3 is incorrect. Down the group, the ease of oxidation increases.	Q36 Answer: B Indium has similar chemical properties to aluminium. Therefore option 1 and 2 are correct.

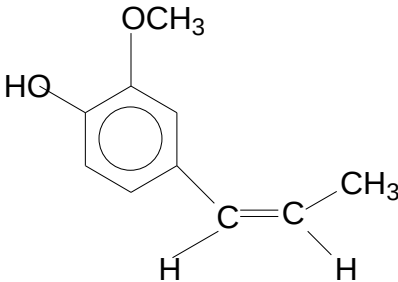
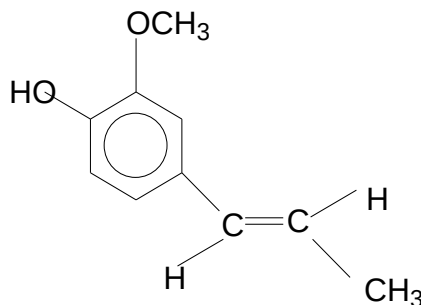
Q37 Answer: A If the temperature is allowed to rise, elimination will occur in propanol to form propene. Therefore option 1 is correct. After elimination, propene will react with HBr to form 2-bromopropane. Therefore option 2 is correct. HBr will react with concentrated sulfuric acid to form Br₂. 2HBr(g) + H₂SO₄(l) → Br₂(g) + 2H₂O(l) + SO₂(g)	Q38 Answer: B Structures for option 1, 2 and 3 respectively:  Option 1 and 2 gives one monochloro-derivative. Option 3 gives 2 monochloro-derivative. Therefore option 3 is incorrect.
Q39 Answer: C Strategy: write out the structural formula and check against the molecular formula (C₄H₉O₂N).  Option 1: does not match the molecular formula. Option 2 and option 3 matches the molecular formula. Therefore they are correct.	Q40 Answer: A For all options: nucleophilic attack occurs at the carbonyl carbon.

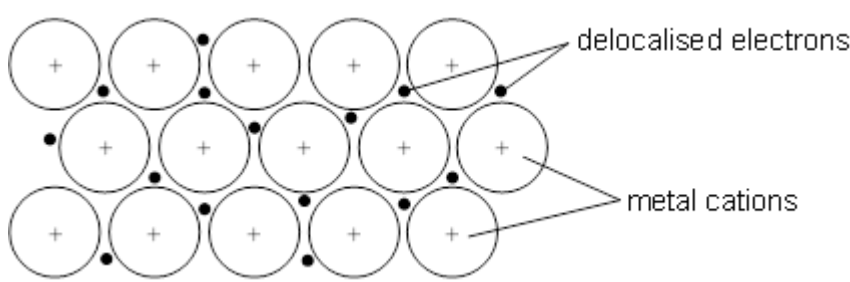
Paper 2 Worked Solutions

Qn	Suggested solutions
2(a)(i)	Reaction (1) is the reaction mainly responsible for raising the temperature of the furnace as it is the most exothermic reaction.
2(a)(ii)	There is increase in the disorderliness in the system for reactions (2) and (4) because there is an increase of 1 mole of gaseous molecules (Δn_{gas} = +1) . There is no change in the amount (in mol) of gaseous molecules (ie. Δn_{gas} = 0) for the other reactions .
2(a)(iii)	$\Delta G = \Delta H - T\Delta S$ $= (-24.8) - (500 + 273)(+0.0158)$ $= \mathbf{-37.0 \text{ kJ mol}^{-1}}$ Increasing temperature will increase the magnitude of TΔS. Hence, ΔG will become more negative and the reaction will be more spontaneous.
2(a)(iv)	Le Chatelier's Principle states that when a system in equilibrium is disturbed, the system will react to counteract the effect of the change until a new equilibrium is reached. When temperature is increased, by Le Chatelier's Principle, the position of equilibrium shifts to the right to absorb the excess heat, favouring the endothermic reaction. Hence, the partial pressure of CO will increase. When pressure is increased, by Le Chatelier's Principle, the position of equilibrium shifts to the left to decrease the total amount (in mol) of gases. Hence, the partial pressure of CO will decrease. (do not need to state whether temperature or pressure would have a greater effect)
2(a)(v)	$\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ $\text{CaO}(\text{s}) + \text{SiO}_2(\text{s}) \rightarrow \text{CaSiO}_3(\text{s})$ Mass of SiO₂ from 1 tonne of ore = $\frac{3.5}{100} \times 1 \times 10^6 = 35000 \text{ g}$ Amount of SiO₂ = $\frac{35000}{60.1} = 582.4 \text{ mol}$ Since SiO₂ ≡ CaO ≡ CaCO₃ Mass of CaCO₃ = 582.4 × 100.1 = 5.83 × 10⁴ g
2(b)(i)	2Fe³⁺ + Sn²⁺ → 2Fe²⁺ + Sn⁴⁺
2(b)(ii)	$E^\ominus_{\text{cell}} = E^\ominus_{\text{red}} - E^\ominus_{\text{oxid}}$ $= +0.77 - (+0.15)$ $= \mathbf{+0.62 \text{ V}}$
2b(iii)	$E^\ominus_{\text{cell}} = E^\ominus_{\text{red}} - E^\ominus_{\text{oxid}}$ $= -0.44 - (+0.15)$ $= \mathbf{-0.59 \text{ V}}$ Since E[⊖]_{cell} < 0, reaction is not feasible under standard conditions.

2(c)(i)	Titration no	1	2	3
	Initial burette reading / cm ³	0.00	21.85	0.00
	Final burette reading / cm ³	21.85	43.25	21.50
	Titre/ cm ³	21.85	21.40	21.50
			Choose titre values ±0.01 cm ³	
Average titre = $\frac{21.40+21.50}{2} = 21.45 \text{ cm}^3$				
Amount of K ₂ Cr ₂ O ₇ = $\frac{21.45}{1000} \times 0.1 = \underline{2.15 \times 10^{-3} \text{ mol}}$				
<i>[Note: Only titre values from titrations 2 and 3 are used as they are within 0.10 cm³ of each other. The first titration gives a rough titre value to see roughly where the end point is.]</i>				
2(c)(ii)	Cr₂O₇²⁻ + 6Fe²⁺ + 14 H⁺ → 6Fe³⁺ + 2Cr³⁺ + 7H₂O			
2(c)(iii)	Amount of Fe ²⁺ in 25 cm ³ = 2.145 x 10 ⁻³ x 6 = 0.01287 mol Amount of Fe ²⁺ in 250 cm ³ = 0.01287 x 10 = 0.1287 mol Mass of Fe in the ore = 0.1287 x 55.8 = 7.18 g % by mass of Fe in the ore = $\frac{7.182}{11.15} \times 100\% = \underline{64.4\%}$			

3(a)(i)	Reagent A: NaOH (<i>*Na not accepted as reaction is v exothermic/ explosive</i>) Reagent B: 		
3(a)(ii)			
3(a)(iii)	Reddish brown bromine solution will be decolourised and white ppt will be observed. Note: while HBr is also formed in the reaction, the small amounts formed would not be sufficient to give a visible observation		
3(a)(iv)			
3(b)(i)	The p-orbital of oxygen overlaps with the π electron cloud of the benzene ring This results in the delocalisation of negative charge on the oxygen into the ring which reduces the intensity of the negative charge on the oxygen atom. This stabilises the phenoxide ion , making phenol more acidic than alcohols. On the other hand, the electron-donating ethyl group in ethanol increases the negative charge on the oxygen atom in CH₃CH₂O⁻ , making the ethoxide ion less stable . This causes ethanol to be less		

	acidic. Note: include both reasons above.
3(b)(ii)	$K_a = 10^{-7.4}$ $[H^+] = \sqrt{10^{-7.4} \times 0.001} = 6.31 \times 10^{-6}$ $pH = -\lg[H^+] = \underline{5.20}$
3(c)	 Cis isomer  Trans isomer

4(a)(i)	Iron, copper and calcium have similar shielding effect because they are in the same period. However, across the period, the nuclear charge increases . This causes electrostatic force of attraction between nucleus and valence electrons to increase , thus the valence electrons to be more strongly attracted to the nucleus in iron and copper than in calcium.
4(a)(ii)	Iron and copper has a smaller atomic radius than calcium. Therefore, iron and copper will have a smaller volume than calcium. Since iron and copper have a larger atomic mass , with a smaller volume, they both have a greater density than calcium.
4(b)(i)	Giant metallic structure Strong electrostatic forces of attraction between the positively charged calcium ion and the sea of delocalised electrons.  note: need to mention specific particles being attracted.
4(b)(ii)	Electrons from the 4s and 3d subshell are delocalised due to their close energy levels as the sea of delocalised electrons from Fe. Only electrons from the 4s subshell are delocalised from Ca. With more electrons delocalised from Fe, it has a stronger metallic bond, requiring more energy to break, leading to a higher melting point for Fe.
4(c)(i)	Copper cables are heavy, causing overhead cables to sag.
4(c)(ii)	Cu: $1s^2 2s^2 2p^6 \underline{3s^2 3p^6 3d^{10} 4s^1}$ Cu²⁺: $1s^2 2s^2 2p^6 \underline{3s^2 3p^6 3d^9}$
4(c)(iii)	Impure copper is obtained through electrolysis. Pure copper is placed at the cathode; impure copper is placed at the anode. The electrolyte is aqueous CuSO₄ solution. Copper in impure copper is oxidised as copper (II) ions and is present in the electrolyte. When electricity is passed through, copper (II) ions from the solution are discharged to form the copper metal. It attaches itself to the pure copper cathode.
4(d)(i)	$Ca(NO_3)_2 \rightarrow CaO + 2NO_2 + \frac{1}{2} O_2$
4(d)(ii)	Copper (II) has a higher charge density because it has a smaller ionic radius than Ca²⁺ . This gives

copper (II) **greater polarising power**, distorting the NO_3^- electron cloud to a greater extent. This **weakens N-O bond to a greater extent**, making it easier to break. Therefore, **$\text{Cu}(\text{NO}_3)_2$ will decompose at a lower temperature.**

5(a) **Structural protein** (e.g keratin), **Enzymatic protein** (e.g pepsin) and **transport** proteins (e.g haemoglobin), **Immunological** (e.g antibodies), **storage protein** (e.g ferritin)

5(b)(i) **Cys- tyr- ile- gln- asn- cys- pro- leu- Gly**
(Start with the N terminus)

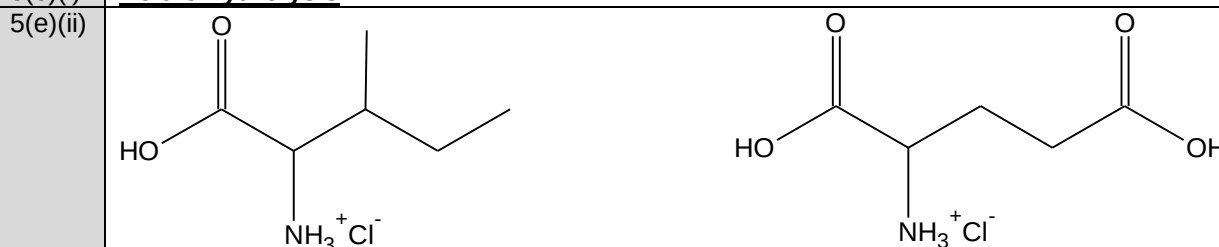
5(b)(ii) The **phenolic group** in the tyr amino acid residue is **acidic** and hence will ionise at pH 7.

5(c) **Tertiary structure** of protein describes the overall 3D shape of protein.

Type of R group interaction	Between R groups of the following amino acid residues
Ionic bonds	Between $-\text{O}^-$ of tyr and $-\text{NH}_2$ of N terminus
Hydrogen bonds	Between tyr, gln and asn
Van der Waals' interactions	Between non polar R groups of leu, pro, ile
Disulfide linkages	Between R groups of Cys and Cys

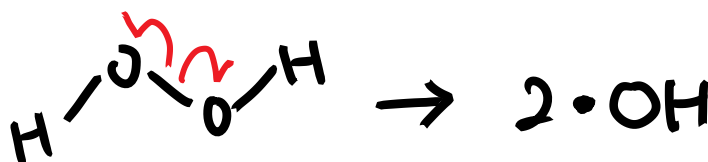
5(d) Hg^{2+} will **disrupt** the **ionic interactions** between ionic R groups, **by forming ionic bond with COO^-** . Hg^{2+} also **react with the $-\text{SH}$ side chains of the cysteine amino acid residues to form precipitates, preventing the formation of disulfide bonds.**
This will result in the protein losing its native conformation.

5(e)(i) **Acidic hydrolysis**

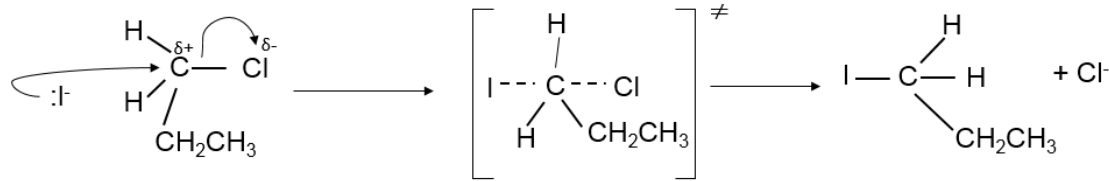


Free radicals are **unstable species** with **unpaired electron on one of the atoms.**

The **O-O bond is weak** and undergoes **homolytic fission** to give rise to **$\bullet\text{OH}$ radicals.**

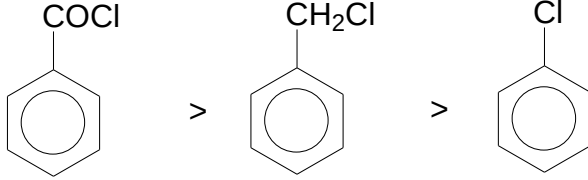
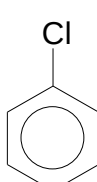


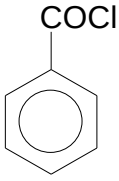
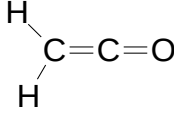
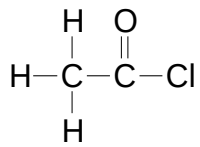
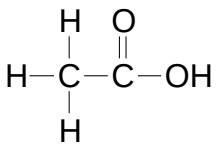
Paper 3 Worked Solutions

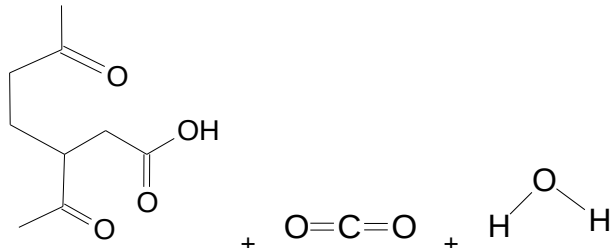
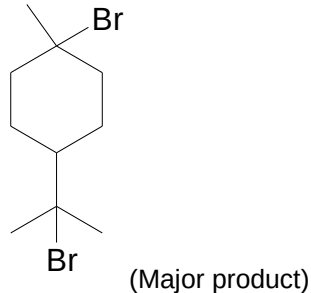
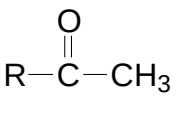
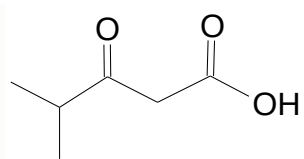
1(a)	- The oxidising power of the halogens can be measured by their standard reduction potentials.<table><tr><td></td><td>$E^\ominus / \text{V}$</td></tr><tr><td>$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$</td><td>+2.87</td></tr><tr><td>$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$</td><td>+1.36</td></tr><tr><td>$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$</td><td>+1.09</td></tr><tr><td>$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$</td><td>+0.54</td></tr></table> - Down the group, reduction potential becomes less positive. - The halogens have lower tendency to be reduced/ oxidizing power of halogen decreases down the group.		$E^\ominus / \text{V}$	$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	+2.87	$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36	$\text{Br}_2 + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	+1.09	$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$	+0.54																				
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1(b)(i)	$2\text{HX}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{X}_2(\text{g})$																														
1(b)(ii)	Down the group, thermal stability of the hydrogen halides decreases. - Down the group, bond length of H–X increases. Bond strength of H–X decreases. Less energy required to overcome the H–X bond. Lower temperature required for thermal decomposition.																														
1(c)(i)	$\text{S}_{\text{N}}2$ mechanism 																														
1(c)(ii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ (propanone) + NaI (propanone) $\rightleftharpoons \text{CH}_3\text{CH}_2\text{CH}_2\text{I}$(propanone) + NaCl (s) As NaCl is precipitated out of propanone solution (given in the question that NaCl is almost insoluble in propanone), $[\text{Cl}^-]$ decreases continuously. By Le Chatelier's principle, the equilibrium position shifts to the right to increase $[\text{Cl}^-]$. Hence the reaction goes almost to completion.																														
1(c)(iii)	- Bromoalkane A undergoes nucleophilic substitution with NaOH (aq) to form alcohol B B undergoes oxidation with $\text{Na}_2\text{Cr}_2\text{O}_7$ to form neutral C $\Rightarrow$ C is a ketone $\Rightarrow$ B is a 2° alcohol and A is a 2° bromoalkane A undergoes nucleophilic substitution with NaOH (aq) to form D $\Rightarrow$ D is a 2° iodoalkane $\frac{M_r \text{ of D} - M_r \text{ of A}}{M_r \text{ of A}} \times 100\% = \frac{170 - 123}{123} \times 100\% = 38.2\%$ Hence A is $\text{CH}_3\text{CHBrCH}_3$, B is $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$, C is CH_3COCH_3 and D is $\text{CH}_3\text{CHICH}_3$																														
1(d)(i)	$K_c = \frac{[\text{I}_3^-]}{[\text{I}_2][\text{I}^-]} \text{ mol}^{-1} \text{ dm}^3$																														
1(d)(ii)	<table><tr><td></td><td>I_2</td><td>+</td><td>I^-</td><td>$\rightleftharpoons$</td><td>I_3^-</td></tr><tr><td>Initial amt/mol</td><td>2.54 / 2(127)</td><td></td><td>(100/1000) x 1.00</td><td></td><td>0</td></tr><tr><td>Change in amt/mol</td><td>- 9.98 x 10⁻³</td><td></td><td>- 9.98 x 10⁻³</td><td></td><td>+ 9.98 x 10⁻³</td></tr><tr><td>Equilibrium amt/mol</td><td>2.00 x 10⁻⁵</td><td></td><td>0.09002</td><td></td><td>9.98 x 100/1000</td></tr><tr><td>Equilibrium [] /mol dm⁻³</td><td>(2.0 x 10⁻⁵) / (100/1000) = 2.00 x 10⁻⁴</td><td></td><td>(0.09002) / (100/1000) = 0.900</td><td></td><td>-</td></tr></table>		I_2	+	I^-	$\rightleftharpoons$	I_3^-	Initial amt/mol	2.54 / 2(127)		(100/1000) x 1.00		0	Change in amt/mol	- 9.98 x 10 ⁻³		- 9.98 x 10 ⁻³		+ 9.98 x 10 ⁻³	Equilibrium amt/mol	2.00 x 10 ⁻⁵		0.09002		9.98 x 100/1000	Equilibrium [] /mol dm ⁻³	(2.0 x 10 ⁻⁵) / (100/1000) = 2.00 x 10⁻⁴		(0.09002) / (100/1000) = 0.900		-
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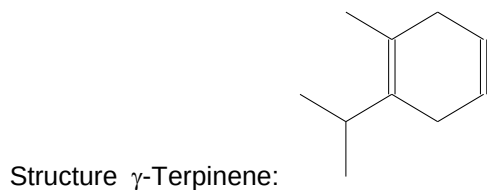
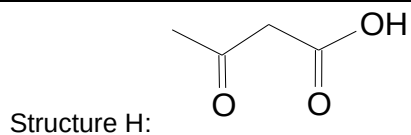
	Hence $K_c = \frac{[I_3^-]}{[I_2][I^-]} = 554.3 = \underline{\underline{554 \text{ mol}^{-1} \text{ dm}^3}}$
1(d)(iii)	$K_c = \frac{[I_3^-]}{[I_2][I^-]}$ $554.3 = \frac{(1.00)}{[I_2](1.00)}$ $[I_2] = 1.804 \times 10^{-3} = 1.80 \times 10^{-3}$
1d(iv)	$I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq) \text{ ----- eqm2}$ <u>Molecular I₂ is much more soluble in hexane</u> because I₂ is able to form favourable VDW interactions with hexane. I⁻ and I₃⁻ ions will remain in the aqueous layer. Hence when hexane is added to the mixture in eqm 2, the <u>[I₂(aq)] will decrease</u>. By Le Chatelier's principle, the <u>equilibrium position for the above reaction will shift left to increase [I₂(aq)]</u>. Hence there will be a decrease in [I₃⁻] and an increase in the [I⁻(aq)] in the new equilibrium.
1d(v)	Initial $E^0_{\text{cell}} = E^0_{\text{red}} - E^0_{\text{ox}} = +0.536 - 0.34 = +0.196 \text{ V}$ $I_3^-(aq) + 2e^- \rightleftharpoons 3 I^-(aq) \quad E^0 = +0.536 \text{ ---- eqm 3}$ When hexane is added to the I₃⁻/I⁻ half cell, [I₂(aq)] will decrease and equilibrium position for the eqm 2 will shift left. Hence, there will be a <u>decrease in [I₃⁻] and an increase in the [I⁻(aq)]</u> in the new equilibrium for eqm 2. In response, by Le Chatelier's principle, the <u>position of eqm 3 will shift left to increase [I₃⁻] and decrease [I⁻] respectively</u>. This result in the <u>E⁰_{red} value for I₃⁻/I⁻ half cell becoming less positive</u>. Hence the <u>new E⁰_{cell} will be less positive</u>.

2a	Bronsted-Lowry base is a species which <u>accepts protons / proton acceptor</u>. <u>Conjugate base</u> is formed when <u>an acid donates its proton</u>; <u>conjugate acid</u> is formed when <u>a base accepts a proton</u>. A <u>conjugate acid-base pair is a pair</u> of species that <u>differs</u> from each other by <u>one proton</u>.
2b(i)	Ammonia acts as a weak <u>base</u>. Ammonia <u>accepts a proton</u> from <u>water, which acts as an acid</u>. This forms the <u>conjugate acid-base pair – NH₄⁺ is the conjugate acid, while OH⁻ is the conjugate base</u>.
2b(ii)	Ammonia acts as a <u>reducing agent</u>. It <u>reduces HNO₂ to N₂</u>, as shown by the decrease in oxidation state from <u>+3 to 0</u> respectively.
2b(iii)	Ammonia acts as an <u>acid</u>. Ammonia <u>donates a proton</u> to <u>NaH, which acts as a base</u>. This forms the <u>conjugate acid-base pair – H₂ is the conjugate acid, while NH₂⁻ is the conjugate base</u>.
2b(iv)	Ammonia acts as a <u>reducing agent</u>. It <u>reduces NaOCl to NaCl</u>, as shown by the decrease in oxidation state from <u>+1 to -1</u> respectively.
2b(v)	Ammonia acts as a <u>nucleophile</u>. The <u>lone pair on N atom of ammonia attacks the partial positive charge of the C on COCl</u>.
2c	<u>Increasing</u> strength of basicity: <u>ethylamine > ammonia > phenylamine</u> In ethylamine, the <u>electron-donating ethyl group increases the availability of lone pair on N</u>, making it <u>more available to accept a proton</u> than in ammonia. In phenylamine, the <u>lone pair of electrons on N is delocalised over the benzene ring</u> due to the <u>overlap of the p orbitals on N</u> and the <u>pi-electron cloud of benzene</u>. This <u>decreases the availability of lone pair on N</u>, making it less available to accept a proton.
2d(i)	4-methylbenzene is <u>more basic</u> than phenylbenzene. In 4-methylbenzene, the presence of the <u>electron-donating methyl group increases the availability of lone pair on N, allowing it to accept a proton more readily</u>, as compared to phenylbenzene with no alkyl groups. Benzylamine is <u>less basic than ethylamine</u>. In ethylamine, the presence of the <u>electron-donating ethyl group increases the availability of lone pair on N, allowing it to accept a proton more readily</u>; as compared to <u>C₆H₅CH₂⁻</u>.
2d(ii)	Bromine water will undergo easy <u>electrophilic substitution with 4-methylphenylamine</u>, but not with benzylamine. As such, the <u>reddish-brown bromine water will be decolourised</u> when reacted with <u>4-methylphenylamine, a white ppt is also formed, and steamy white fumes of HBr is formed</u> as by-product; but <u>no colour change</u> is observed with benzylamine.
2d(iii)	Route 1: <i>Note: All state symbols, conditions (eg, heat in sealed tube is not the same as reflux) and quantities (limited or excess) are imp!</i>

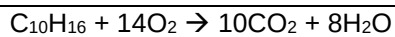
3(a)	<u>$P_4O_6(s) + 6H_2O(l) \rightarrow 4H_3PO_3(aq)$</u> P_4O_6 reacts <u>vigorousl</u> with water to for an <u>acidic solution of pH 3 (pH 2 is acceptable)</u>. Al_2O_3 does not react with water due to <u>very stable lattice structure</u> <u>→ very high ΔH_{latt}</u> <u>∴ insoluble</u>
3(b)	Oxide W has a <u>giant ionic structure</u>. It has high melting point as a <u>large amount of energy</u> is needed to <u>break the strong ionic bonds</u>. It can conduct electricity in the molten state due to the <u>presence of mobile ions</u>. It has very little reaction with water due to very stable lattice structure/very high ΔH_{latt}. Oxide X has <u>giant molecular structure with covalent bonds between the atoms</u>. It has high melting point as a <u>large amount of energy is needed to break the strong covalent bonds between the atoms</u>. It cannot conduct electricity in the molten state due to the <u>absence of mobile ions or mobile electrons</u>. Oxide Y has a <u>giant ionic structure</u>. It has high melting point as a <u>large amount of energy is needed to break the strong ionic bonds</u>. It can conduct electricity in the molten state due to the <u>presence of mobile ions</u>. Oxide Z is a <u>simple molecular compound with intermolecular Van der Waals forces of attraction and covalent bonds between the atoms in the molecules</u>. It has low melting point as a <u>small amount of energy is needed to break the weak Van der Waals forces of attraction</u>. It cannot conduct electricity in the molten state due to the <u>absence of mobile ions or mobile electrons</u>.
3(c)(i)	From the data booklet: $SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e^-$ $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ <u>$2MnO_4^- + 5SO_2 + 2H_2O \rightarrow 2Mn^{2+} + 5SO_4^{2-} + 4H^+$</u>
3(c)(ii)	<u>Colourless</u> solution turn <u>pink</u>
3(c)(iii)	Amount of $MnO_4^- = \frac{20.5}{1000} \times 0.02 = 4.1 \times 10^{-4} mol$ Amount of $SO_2 = \frac{4.1 \times 10^{-4}}{2} \times 5 = 1.025 \times 10^{-3} mol$ Concentration of $SO_2 = \frac{1.025 \times 10^{-3}}{25} \times 1000 = 0.0410 mol dm^{-3}$
3(d)	An acidic buffer is formed. $pH = pK_a + \lg \frac{[salt]}{[acid]} = -\lg(6.3 \times 10^{-5}) + \lg \frac{(0.03)}{(0.01)} = 4.68$
3(e)	  : The <u>p orbital of the chlorine atom can overlap with the π-electron cloud of the benzene ring</u>. As a result, the <u>lone pair of electrons on the halogen atom can delocalise into the benzene ring</u>. This results in <u>partial double bond character</u> in C-X bond and the <u>strengthening</u> of the C-X bond. Hence, the halogen atom is difficult to substitute.

	 : The carbonyl C atom has two very electronegative atoms (O and Cl) bonded to it. Both O and Cl exert electron-withdrawing effects and hence the carbonyl C atom has a large partial positive charge making the compound more susceptible to nucleophilic substitution.
3(f)	E:  F:  G: 

4(a)	Optical isomerism. Two optical isomers
4(b)	
4(c)	 (Major product)
4(d)	- H and J undergo acid-base/ neutralisation reaction with Na_2CO_3 $\Rightarrow$ H and J each contains a -COOH group - Both H and J undergoes condensation reaction with 2,4-DNPH to give hydrazones $\Rightarrow$ H and J are carbonyl compound - Both H and J do not undergo oxidation reaction with Fehling's solution $\Rightarrow$ H and J are ketones / aromatic aldehyde - H undergoes oxidation with alkaline aqueous iodine while J does not. $\Rightarrow$ H contains  structure - K undergoes oxidation to form J. Structure J: 



4(e)(i)



4(e)(ii)

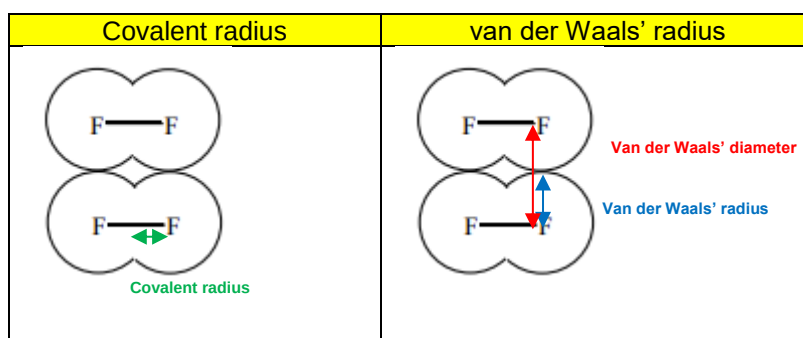
$$\begin{aligned} \Delta H_c \text{ of limonene} &= 16\text{BE}(\text{C}-\text{H}) + 8\text{BE}(\text{C}-\text{C}) + 2\text{BE}(\text{C}=\text{C}) \\ &+ 14\text{BE}(\text{O}=\text{O}) - 10 \times 2 \times (805) - 8 \times 2\text{BE}(\text{H}-\text{O}) \\ &= \mathbf{-5936 \text{ kJ mol}^{-1}} \end{aligned}$$

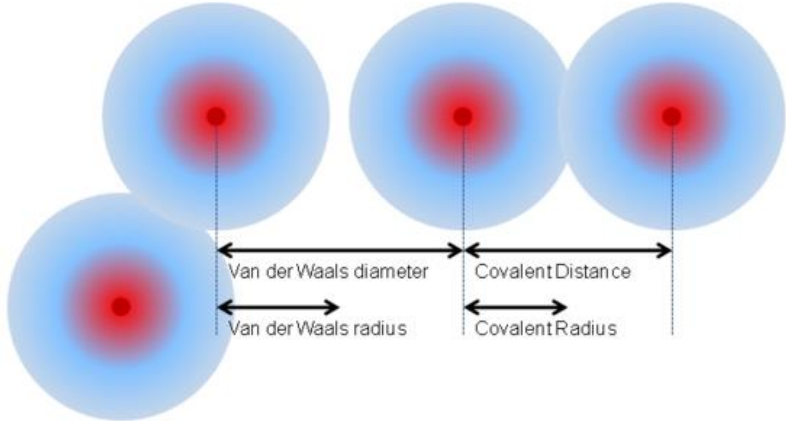
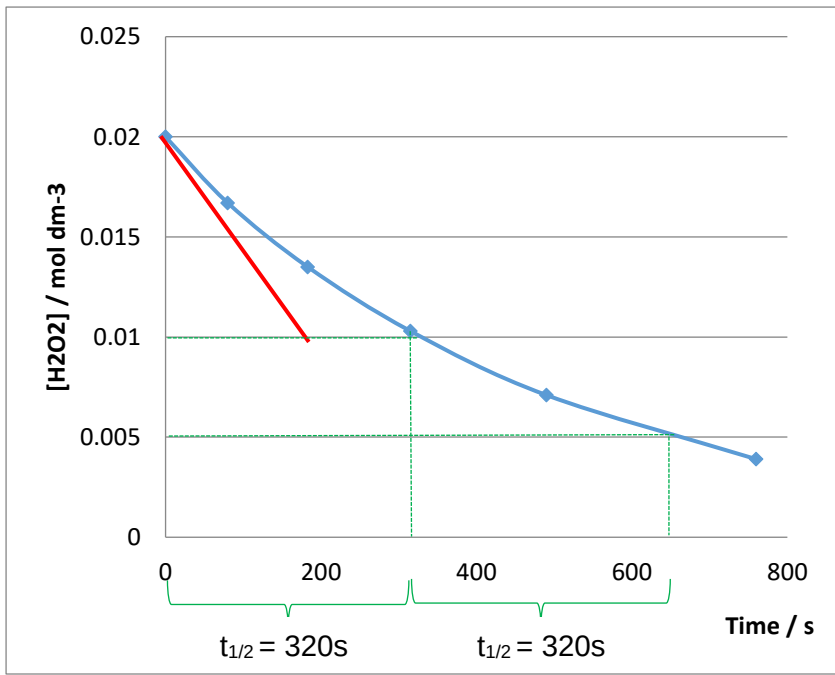
4(e)(iii)

$$\begin{aligned} \Delta H_c \text{ of limonene} &= -5936 / (12 \times 10 + 16) \\ &= \mathbf{-43.6 \text{ kJ g}^{-1}} \end{aligned}$$

This is **less exothermic** compared to that of conventional diesel fuel.

Understand this before attempting 5(a):



5(a)(i)	Silicon has a giant molecular structure with strong covalent bonds between Si atoms . Covalent radius of 0.117nm refers to half the distance between the nuclei of Si atoms in a covalent bond . (not Si – Si distance)
5(a)(ii)	Argon exists as a noble gas with fully filled subshells and does not form covalent bonds with other Ar atoms. Therefore it exists as a single atom.
5(a)(iii)	van der waals' radius of chlorine will be smaller than that of argon. This is so as as the VDW forces of attraction between chlorine molecules is stronger than that between argon atoms as the former has a larger number of electrons, resulting in larger polarisable electron cloud. This stronger VDW force will result in a shorter distance between the molecules and thus smaller van der waals radius Fyi:  Figure: Van der Waals radius and Covalent radius (illustrated by Heesung Shim)
5(b)(i)	Peroxide anion: O_2^{2-} 
5(b)(ii)	$Na_2O_2 + 2H_2O \rightarrow H_2O_2 + 2NaOH$ (Na_2O is not accepted as a product because qns states that an alkaline solution is formed)
5(c)(i)	The order of reaction with respect to a given reactant is defined as the power to which the concentration of that reactant is raised to in the rate equation .
5(c)(ii)	Half-life of a reaction is the time taken for the initial concentration of a reactant to decrease to half its original value .
5(d)	
5(d)(i)	Constant half-life of 320s is observed. Reaction is first order with respect to H_2O_2 . (with relevant

	working from graph)
5(d)(ii)	Initial rate = $ (0.02 - 0.01) / (0 - 190) = \underline{5.26 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}}$ (with relevant working from graph)
5(d)(iii)	Keeping $[I^-]$ constant, when $[H^+]$ increases by 4/3 times, rate also increases by 4/3 times. Order of reaction with respect to H^+ is one . $\frac{8.4 \times 10^{-5}}{2.1 \times 10^{-5}} = \frac{k \times (0.4)(0.2)^x}{k \times (0.2)(0.1)^x}$ $X = 1$ Order of reaction with respect to I^- is one .
5(d)(iv)	Rate = k $[H_2O_2]$ $[I^-]$ $[H^+]$ $k = 8.4 \times 10^{-5} / (0.4 \times 0.2 \times 0.02) = \underline{0.0525 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}}$
5(e)(i)	LiAlH₄ in dry ether, rtp / NaBH₄ in ethanol, rtp
5(e)(ii)	K₂Cr₂O₇ in H₂SO₄(aq) Distillation is to be done immediately to obtain propanal.

END 2013